TaskFlow-Travel: Imitation-Learned Coordination for Multi-Agent Travel Replanning under Evolving User Constraints

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Abstract

Travel planning is rarely completed in one turn. After seeing an itinerary, a user may lower the budget, change the dates, add another traveler, remove a hotel requirement, or reverse the priority between cost and convenience. Current LLM agents can respond fluently to these updates but may forget earlier constraints or regenerate unaffected parts of the plan. We propose TaskFlow-Travel, a multi-agent travel assistant with an imitation-learned coordinator for replanning under evolving user constraints. The coordinator will learn from verified expert trajectories to decide whether to preserve the current itinerary, ask for clarification, call a specialized travel agent, revise a local component, or rebuild the plan globally. We will derive multi-turn tasks from the public TravelPlanner and FlexTravelBench resources and compare a single agent, a static multi-agent workflow, and our learned multi-agent system. Evaluation will measure final constraint satisfaction, retention of earlier requirements, unnecessary replanning, and tool-use cost.

1 Idea and Motivation

Large language models have made tool-using travel assistants plausible, but producing a valid multi-day itinerary remains a constrained planning problem. A plan must jointly satisfy dates, transportation, opening hours, location, room rules, cuisine preferences, and a total budget. TravelPlanner provides public tools, records, and reference plans for evaluating these requirements [1]. Flex-TravelPlanner extends this setting by revealing constraints over multiple turns and shows that strong models often accommodate a newly introduced preference at the expense of an existing global constraint such as budget [2].

Our project focuses on a more realistic interaction: the user's requirement may not only be added, but also deleted, replaced, or reprioritized. For example, the budget may change from $3,000 to $2,200, a downtown-hotel preference may be removed, or convenience may become more important than price. A useful agent should identify what the change invalidates, preserve unaffected decisions, and replan only where necessary.

1.1 Why It Is Useful

Recomputing an entire itinerary after every update is expensive and unstable. It may discard choices the user already approved, introduce new violations, and repeat unnecessary searches. A reliable replanning assistant could reduce interaction cost while making plan revisions easier to understand. The same coordination problem also appears outside travel whenever an agent must maintain a long plan while requirements change.

1.2 Why It Is Interesting

The task exposes a tension between stability and adaptability. An agent must remember valid earlier decisions, but it must also change enough of the plan to satisfy the new request. Recent multi-agent systems improve travel planning through explicit constraint tracking, checking, search, and hierarchical decomposition [3, 4]. However, a fixed workflow may not know whether a particular update requires a small repair or a global restart. We therefore study whether this coordination decision can be learned from demonstrations instead of being completely hand-coded.

2 Potential Solution

2.1 Evolving-Constraint Task Construction

We will build on the public TravelPlanner and FlexTravelBench data. Original queries will be divided into train, validation, and test sets before new variants are generated, preventing similar versions of one trip from crossing splits. Each itinerary will receive one or more controlled updates from four categories: addition, deletion, replacement, and priority change. Test cases will include unseen combinations, change orders, and natural-language paraphrases. A deterministic checker will label whether the final plan satisfies both the current request and all still-active earlier constraints.

2.2 Multi-Agent Planning System

TaskFlow-Travel will include a Constraint Tracker, Transportation Agent, Accommodation Agent, Activity and Dining Agent, Plan Composer, and Constraint Verifier. These roles may share the same underlying LLM; multi-agent refers to modular responsibilities rather than different model families. The system will maintain a structured constraint state and cache earlier tool results so that unchanged information can be reused.

2.3 Imitation-Learned Coordinator

The coordinator observes the conversation, current itinerary, constraint change, verifier output, and available cached information. It selects structured actions such as KEEP_PLAN, ASK_CLARIFICATION, SEARCH_TRANSPORTATION, SEARCH_ACCOMMODATION, REPLAN_LOCAL, REPLAN_GLOBAL, VERIFY, and STOP. Training trajectories will be generated by a stronger teacher combined with the deterministic verifier; only verified successful trajectories will be retained. We will use behavior cloning to train a small coordinator with supervised state-action pairs. This keeps the deep-learning component manageable while directly targeting the sequential decision problem. Prior work has shown that imitation learning can improve tool-using language agents, but it has not resolved our specific question of learned multi-agent coordination under genuine constraint revision [5].

2.4 Experimental Design

We will compare three systems: (1) a Single-Agent baseline with access to all tools, (2) a Static Multi-Agent system with a fixed rule- or prompt-based coordinator, and (3) Multi-Agent + IL with the learned coordinator. Wherever possible, the systems will use the same base LLM, databases, tools, and total tool-call budget. Primary metrics will include Final Pass Rate, New Constraint Satisfaction, Constraint Retention Rate, Revision Success Rate, the fraction of unaffected itinerary components preserved, number of tool calls, and token cost. Results will be reported separately for each change type and interaction length.

3 Expected Outcomes

We expect to produce: (1) a reproducible set of multi-turn travel tasks containing additions, deletions, replacements, and priority changes; (2) a modular travel-planning prototype; (3) an imitation-learned coordination policy; and (4) a controlled evaluation of accuracy, plan stability, and efficiency. Our hypothesis is that learned coordination will improve constraint retention and avoid unnecessary global replanning compared with the two baselines. This hypothesis will be rejected if the learned system does not outperform the static multi-agent workflow under a comparable budget, fails on held-out change types, or provides gains only through additional tool calls. If simple rules perform equally well, the result will show that learning is unnecessary for this task boundary.

4 Rough Timeline

- **Weeks 1-2:** Finalize the constraint taxonomy, reproduce the public benchmark pipeline, and implement the verifier.
- **Weeks 3-4:** Generate evolving-constraint scenarios, validate labels, and create leakage-safe splits.
- **Weeks 5-6:** Implement and evaluate the Single-Agent and Static Multi-Agent systems.
- **Weeks 7-9:** Generate verified demonstrations, train the imitation-learning policy, and integrate it with the agents.
- **Weeks 10-11:** Run controlled comparisons, ablations, cost analysis, and failure analysis.
- **Week 12:** Complete the report, presentation, reproducibility package, and demonstration.

References

[1] J. Xie, K. Zhang, J. Chen, T. Zhu, R. Lou, Y. Tian, Y. Xiao, and Y. Su. 2024. TravelPlanner: A Benchmark for Real-World Planning with Language Agents. ICML 2024.

[2] J. Oh, E. Kim, and A. Oh. 2025. Flex-TravelPlanner: A Benchmark for Flexible Planning with Language Agents. arXiv:2506.04649.

[3] J. Choi, J. Yoon, J. Chen, S. Jha, and T. Pfister. 2025. ATLAS: Constraints-Aware Multi-Agent Collaboration for Real-World Travel Planning. arXiv:2509.25586.

[4] T. V. Bui, W. Li, and Y. Liu. 2026. HiMAP-Travel: Hierarchical Multi-Agent Planning for Long-Horizon Constrained Travel. arXiv:2603.04750.

[5] K. Zhang et al. 2026. Agent Learning via Early Experience. arXiv:2510.08558.